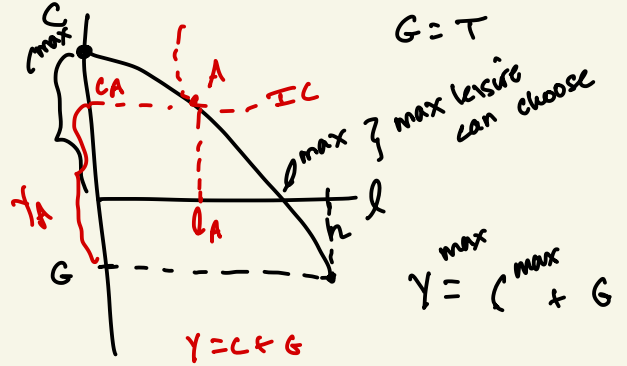




A is an eqm:

1. $MRS = MRT$

2. Highest IC possible

$\frac{MU(C)}{MU(L)} = MRT = MPN$

ch 4

$MRS = w(1-t)$
 $= MPN(1-t)$

only happens if
 G is paid for with
 T rather than
 t

$\neq MPN$

if $t > 0$

A is efficient (no wasted resources)
Parco Optimum

The CHS model is referred to as "Solving the Social Planner's Problem"

"benevolent dictator"

finds A and then

tells people

$$C = C_A$$

$$l = l_A$$

$$N = N_A$$

$$Y = Y_A$$

This equilibrium implies a certain MPNA at N_A and a certain W_A

Social Planner eqm = Pareto Optimum = Efficient Equilibrium



2 Welfare Theorems

First Welfare Theorem: In some cases a market equilibrium (competitive)

is efficient (same as the Social Planner's equilibrium)

It "holds" if perfect competition holds: all price-takers, no externalities, no proportional income taxes, perfect capital markets, no unemployment

Second Welfare Theorem: In some cases a Social Planner's equilibrium will be the same as the market (competitive) equilibrium

→ same circumstances

3 Things in Ch 5

1. Define efficiency (Welfare Theorems)

2. Solve for w
Social Planner's choice
of L and N implies
 MPN which implies w

3. Use the model to understand
sources of business cycles

★ goal: explain why
 C, N, Y, Z, w
are all pro-cyclical over
business cycle

From Ch 4 math: we have to
assume quasi-linear utility
(substitution effects dominate
income effects)

Try: Δ in preferences
 Δ in G (Δ in τ)
 Δ in Z (TFP)

e.g. # 1 Δ in preferences toward
C away from L

$$\downarrow \text{MRS} = \frac{\text{MUC}_D}{\text{MUC}_C} \uparrow$$

In eijn

$$\downarrow \text{MRS} = \downarrow \text{MPN} = w \downarrow$$

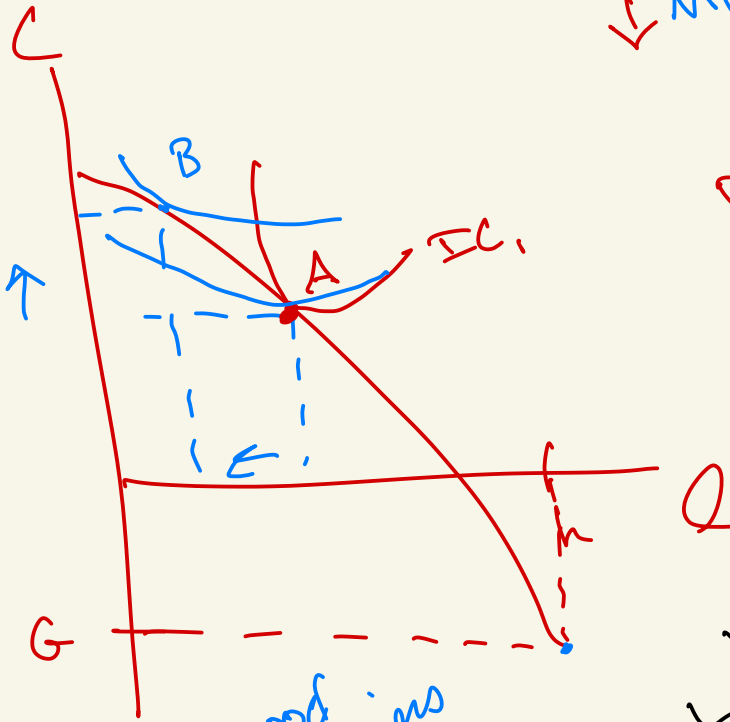
latter
part of
PPF

↳ C↑

$$V = C + G$$

$\gamma \uparrow$ as $w \downarrow$

not what we
observe in business
cycle data



not a good
source of business
cycles in time

e.g. #2

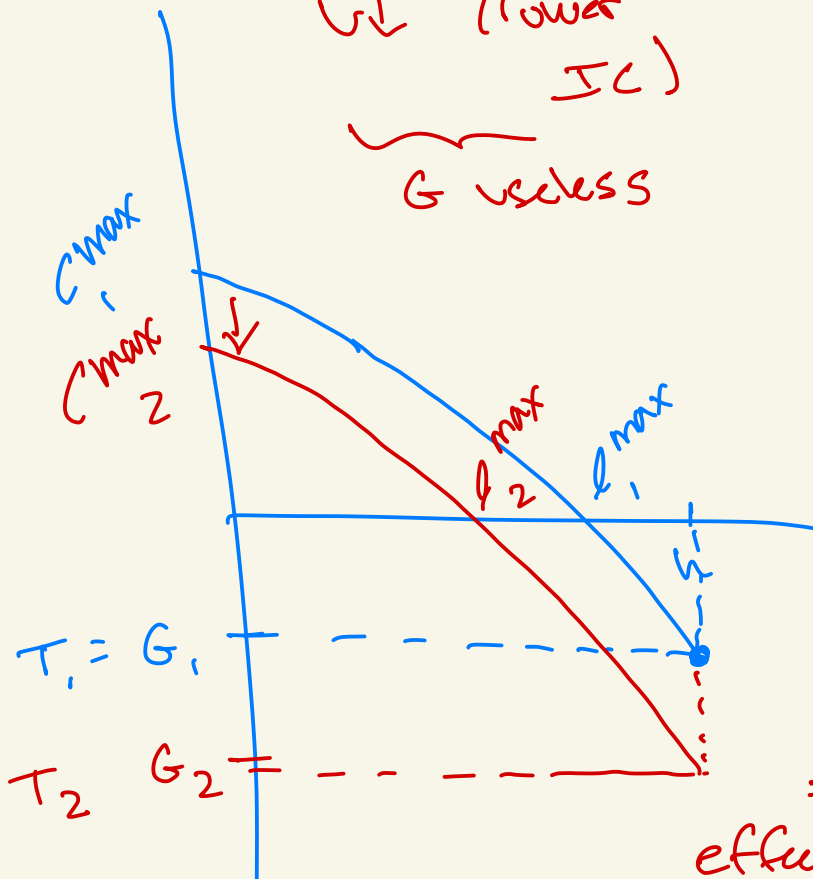
if

$$\Delta G > 0 \Rightarrow$$

what

happens
in the
model?
does it
look like
the cyclical
data?

$\downarrow$ (lower
IC)
 $\underbrace{\hspace{2cm}}$
G useless



$G \uparrow T \uparrow$
 $l \downarrow l^{\max} \downarrow$

$\underbrace{\hspace{2cm}}$
no Δ in
slope

$\Rightarrow$ pure income
effect (like in ch4)

income $\downarrow$ after taxes $\underbrace{C \downarrow l \downarrow}_{\text{normal goods}}$

$\underbrace{\uparrow \uparrow}$

NT MPN $\downarrow \therefore w \downarrow$
 $\nwarrow \nearrow$ } unlike cyclical
ch oh data

e.g. #3 $\Delta TFP (\Delta Z) > 0$

From both income
and sub effects
 $C \uparrow$

$Z \uparrow \} Y \uparrow$

because
 $Y = ZK^2N^\beta$
 $\uparrow \quad \uparrow$

income effect

$L \uparrow$ $C \uparrow$

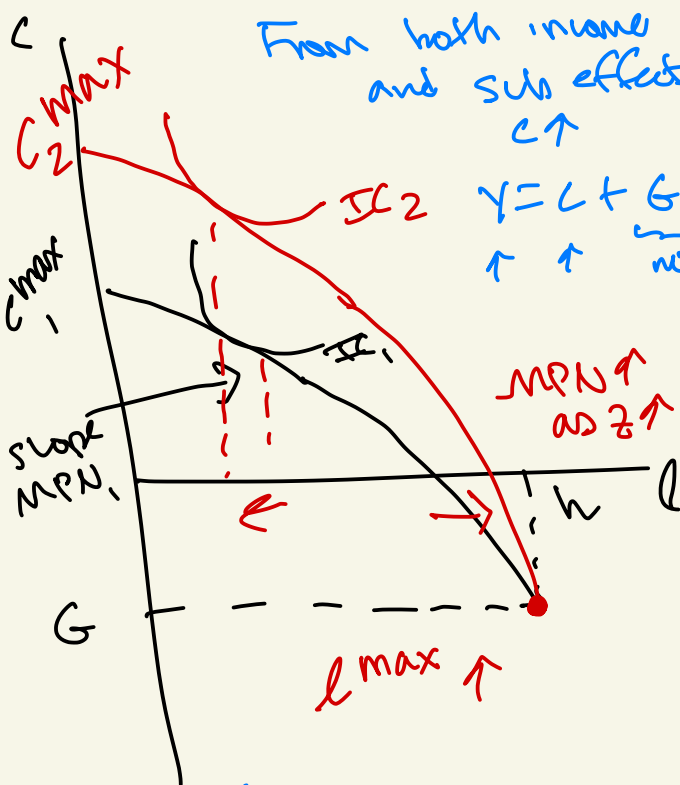
$Z \uparrow$ causes
 $MPN \uparrow$

because
 $MPN = \frac{\partial Y}{\partial N} = ZK^2N^{\beta-1}$
 $\uparrow \quad \uparrow$
 $\quad \quad Z \uparrow$

$L \downarrow$ $N \uparrow$
 $C \uparrow$ $\left\{ \begin{array}{l} W \uparrow \text{ as } Y \uparrow \\ MPN \uparrow = MRS \uparrow \end{array} \right.$

$\Delta L ? \Delta N ?$ $N \uparrow$ like
in data if

~~substitution~~ effects
dominate



* utility $\uparrow$ as
budget constraint
(PPF) $\uparrow$

Productivity
shocks work
to explain
business
cycles

Summary: TFP really important
 $\Rightarrow$ explains long run growth
and is at least very important
for understanding cycles

measuring TFP = unobservable
how good management is
how useful is the internet?
software?

AI?
rainfall, climate shocks

We measure it as a residual of
output minus observable
K and N inputs

$$Y = Z K^{\alpha} N^{\beta}$$

$$\frac{Y}{K^{\alpha} N^{\beta}} = Z \quad \left. \vphantom{\frac{Y}{K^{\alpha} N^{\beta}}} \right\} \begin{array}{l} \text{estimate} \\ \text{of} \\ \text{TFP} \end{array}$$

$$Z = \frac{Y}{K^\alpha N^\beta}$$

$$\ln Z = \ln Y - \ln [K^\alpha N^\beta]$$

Solow Residual

Solow

Problems with Solow residual:

1. The Y, K, N, α, β are collected with error and Solow residual has as many errors.
2. Likely some missing factor inputs (human capital, electricity, other quality adjustments)
3. Production function may change over time